



IDS Pulse — Complete Function & Feature User Guide

This guide provides a detailed, step-by-step breakdown of every single tab, button, field, and workflow in both the IDS Pulse Web Portal and the Field Representative Mobile App.

Part 1: Field Operations (Mobile App / Simulator)

The mobile app is used by Quality Representative Engineers (QREs) on the plant floor. It functions inside the phone simulator preview in the layout.

1. Pin-Code Security Access (Gate)

- Who Can Access: All system users (Reps, Admins, Accountants, Customers) using their unique passcodes.
- Step-by-Step Instructions:

1. Find the mobile device simulator frame in the layout.
2. Enter your 5-digit numerical passcode on the keypad (e.g., Hugo's default passcode is `11223`, Nabil's is `44556`, Shahroz's master override is `55443`).
3. Click the cyan Unlock Terminal button.
4. If correct, you will be authenticated and redirected to your role-specific mobile home screen. If incorrect, an authentication error will flash.

2. Shift Tracker (Clock-In / Clock-Out)

- Who Can Access: Field Representatives (`qre`).
- Step-by-Step Instructions:
- Starting Your Shift:

1. Upon arriving at the plant, locate the green Start Shift button at the top of the mobile home screen.
2. Tap Start Shift.
3. The system captures the current timestamp and logs your GPS coordinates (mocked location).
4. An automated notification registers in the background, updating Colleen (Accountant) and Shahroz (Owner) that you are active at the plant.

- Ending Your Shift:

1. At the end of your workday, tap the red End Shift button.
2. A popup checklist will appear asking you to confirm:
 - Hours worked today.



- Mileage driven (kilometers).
- Walked area inspection status.

3. Tap Confirm & End Shift to log out and submit your timesheet entry for Colleen's payroll approval.

3. Reporting a Quality Incident Defect

- Who Can Access: Field Representatives (`qre`) and Leads.
- Step-by-Step Instructions:

1. On the home screen, tap the New Quality Incident button. This opens a 5-step wizard:

2. Step 1: Part Info & Barcode Scan:

- Select a part description from the dropdown list or type in a custom Part Number (e.g., `12V-BAT-309`).
- To scan a part barcode, tap the Camera/Scanner Icon. This opens the simulator scan view. Point your simulated lens at the part label and press Capture Scan to parse the serial code automatically. Click *Next*.

3. Step 2: Location & Supplier:

- Select the Supplier Client (e.g., *Auto Kabel* or *Brose*).
- Select the Plant Location where containment is active (e.g., *Mercedes Tuscaloosa*). Click *Next*.

4. Step 3: Supplier Specific Fields:

- If you selected Magna AutoSystems, specific fields appear (PRR Quality Class, Containment Level). Fill these out.
- If you selected Auto Kabel, fill out the Standard Wire Gauge (SWG) and insulation checks. Click *Next*.

5. Step 4: Image Capture & Annotation Canvas:

- Tap Take Photo to simulate capturing a picture of the physical part defect.
- Once the mock photo is loaded, tap Draw Annotations.
- A canvas overlay will appear. Click and drag with your mouse/finger to draw red containment circles or arrows directly over the defect (e.g., circling a cable fracture or structural dent).
- Tap Save Markups to attach the annotated photo. Click *Next*.

6. Step 5: Review & Submit:

- Inspect the summary of your defect report.
- Tap Submit Defect Incident. The incident will instantly write to local cache, sync with the Supabase database, and trigger a manager alert.

4. Logging Sorted & Reworked Components

- Who Can Access: Field Representatives (`qre`).
- Step-by-Step Instructions:

1. Tap the Rework Log button on the home screen.

2. Select the Supplier Client and the Plant Location.

3. Input the target Part Number.



4. Fill in two key numerical fields:

- Total Parts Sorted: The total number of components inspected during the shift.
- Total Parts Reworked: The number of components that had defect issues repaired/fixed.

5. Click Submit Rework Log to save the entry.

5. Logging Travel Mileage & Reimbursable Expenses

- Who Can Access: Field Representatives (`.qre`).
- Step-by-Step Instructions:

1. Tap Log Expense on the home screen.

2. Select the Reimbursement Category (e.g., *Fuel, Tolls, Meals, Parking, Safety Gear*).

3. Enter the total Amount spent (e.g., `45.50`).

4. Tap Capture Receipt Photo to take a picture of the paper receipt invoice.

5. Enter a brief justification in the Description box (e.g., "Toll gate fee during Mercedes Tuscaloosa quality containment check").

6. Tap Log Expense Claim to submit to Colleen's audit queue.

Part 2: Accountant Workspace (Colleen's Portal)

Access Colleen's workspace by navigating to the Timesheets & Mileage tab on the sidebar. Under this view, you have 5 specialized financial sub-tabs:

1. Log Hours & Expenses Sub-Tab

- Function: Audits representative submissions.
- Step-by-Step Instructions:

1. View the table of pending hours and expense claims.

2. Hover over or click on any expense attachment to display the uploaded Receipt Photo.

3. Click the green Approve button to confirm the hours/expense for billing/payroll, or click red Reject to deny the claim.

2. Invoicing Control Sub-Tab

- Function: Auto-generates billing invoices for clients based on approved hours.
- Step-by-Step Instructions:

1. Select the Supplier Client (e.g., *Auto Kabel*) from the dropdown.



2. Select the starting and ending dates for the billing period.
3. The system automatically pulls all approved timesheets for that client and calculates the billing amount based on the default rates.
4. The screen will render a professional IDS Invoice Preview including:
 - Total Sort/Rework Hours billed at the client's rate (e.g., 85 hrs @ \$35.00/hr).
 - Applied tax rates.
 - Invoice reference ID.
5. Click Generate Invoice to save the invoice history and display a printable view.

3. Rep Payroll Sub-Tab

- Function: Reviews representative pay summaries and processes payroll.
 - Step-by-Step Instructions:
1. The screen displays a summary table of unpaid hours and mileage reimbursement for each Representative.
 2. Mileage Payout Calculation: Travel mileage submitted during shifts is reimbursed at a standard rate of \$0.68 per kilometer.
 3. The total payroll payout is shown as: ``(Hours worked * Rep Hourly Pay Rate) + (Kilometers * $0.68) + Approved Expenses``.
 4. Click Process Payroll Transfer to mark the logs as "paid" and generate payroll logs.

4. Clients & Rates Sub-Tab (Setup & Onboarding CRUD)

- Function: Provides a consolidated setup portal to manage business entities.
- Step-by-Step Instructions:
- Manage Customers:
- Input the Company Name, primary contact name, email, and invoice schedule (weekly, bi-weekly). Click Onboard Customer.
- Manage Locations:
- Select parent supplier, enter Plant Name (e.g., *Mercedes Tuscaloosa*), address, default QRE representative, standard weekly target hours, and default bill rate (e.g., `\$35.00`). Click Map Location & Rates.
- Onboard Reps:
- Enter the representative's full name, email, hourly pay rate, and custom security passcode. Click Create Representative.

5. Bulk Hours Entry Sub-Tab

- Function: Bypasses mobile app logging to make quick manual entries.
 - Step-by-Step Instructions:
1. If a rep forgot to log hours or you need to adjust rates manually, select the Bulk Hours Entry sub-tab.



2. Enter the Date, select the target Representative, choose the Supplier Client and Plant Location.
3. Enter the total Hours worked and Mileage (km).
4. Click Add Bulk Log Entry to push the timesheet directly into the database.

Part 3: Operations & Quality Management (For Admins & Leads)

Managers use these panels to coordinate field logistics and monitor manufacturing metrics:

1. Incident Defects Feed

- Function: Reviews representative quality submissions and publishes them.
- Step-by-Step Instructions:
 1. Click Incident Defects Feed on the left navigation sidebar.
 2. View list of defect logs (shows part serials, description, and annotated defect images).
 3. Defect logs are flagged as *Internal Only* on submission.
 4. Click the green Publish to Customer button on verified quality issues. This sends the defect log directly to the Customer Portal.

2. Visual Defect Matrix (Heatmap)

- Function: Visually tracks containment defects across plants.
- Step-by-Step Instructions:
 1. Click Visual Defect Matrix on the sidebar.
 2. View the grid: rows display Supplier Clients and columns display Plant Locations.
 3. Cell color intensity ranges from dark gray/blue (low defects) to bright orange/red (high defect rates).
 4. Hover your mouse pointer over any cell to see the tooltips displaying the exact defect counts.

3. Daily Tasks Planner & Rep Task Assignment

- Function: Dispatches lists of tasks and location assignments to Reps.
- Step-by-Step Instructions:
 1. Click Daily Tasks Planner on the sidebar.
 2. Filter by representative to check their current daily task completion status.
 3. To assign a new task:
 - Enter the Task Description (e.g., "Sort incoming battery racks on Line 3").
 - Select the target Representative.



- Choose the Plant Location.
- Click Assign & Dispatch Task. The task is written to the representative's task list in their mobile app instantly.

4. Projects Registry

- Function: Configures quality containment projects.
- Step-by-Step Instructions:

1. Click Projects Registry on the sidebar.
2. To create a new project, click Create New Project.
3. Fill in: Project number (e.g., `PRJ-FOR-12`), Supplier client, containment description, location/plant, currency, billing rate, and rep pay rate.
4. Click Submit Project to persist it in the Supabase `projects` table.

5. User Directory & Passcode Reset

- Who Can Access: Admins, Shahroz.
- Step-by-Step Instructions:

1. Click User Directory on the sidebar.
2. View user details including names, roles, and current passcodes.
3. To update a passcode, click the Edit button next to a user, type the new 5-digit number, and click Save.

6. Launch Roadmap (Shahroz Dashboard)

- Who Can Access: Shahroz only.
- Step-by-Step Instructions:

1. Click Launch Roadmap on the sidebar.
2. View development checklist items (Express server backend, file storage setup, app store submission pipelines).
3. Click checkbox items to log project progress.